

NAME : SHAH FAISAL

ROLL NO : 23I-0058

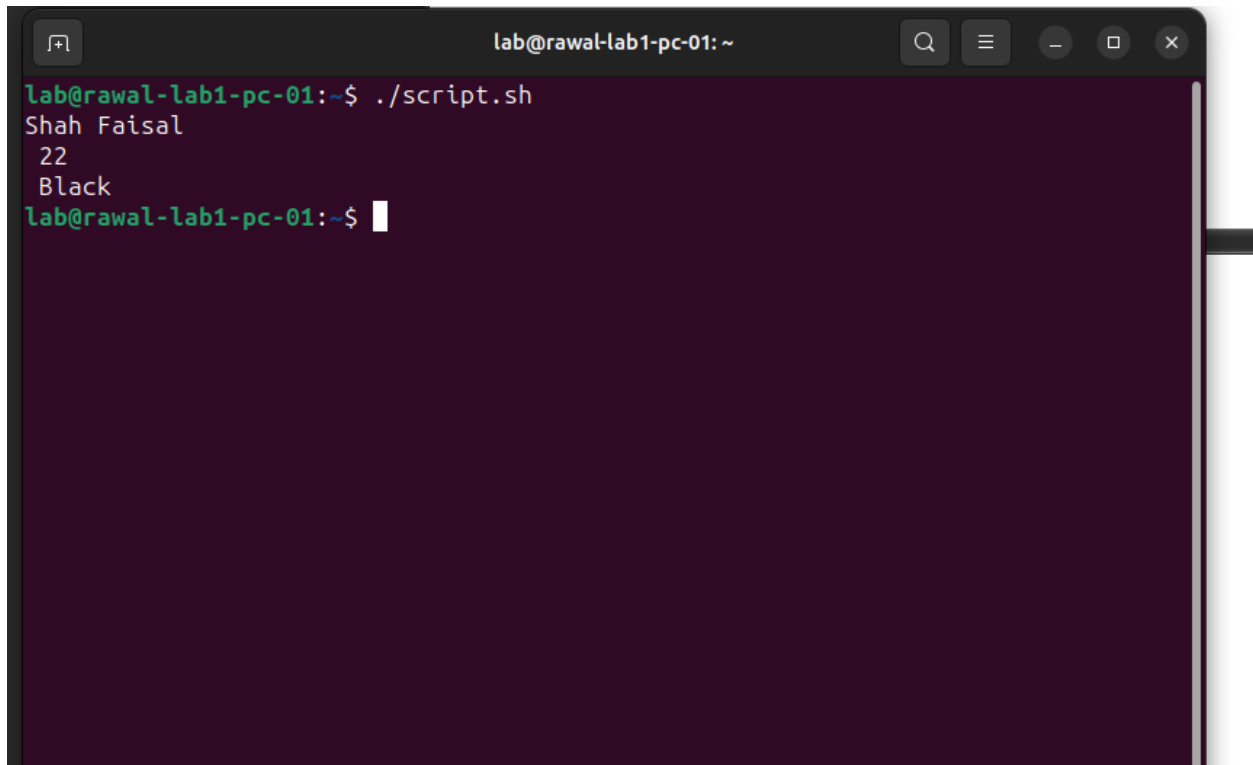
SECTION : BSE – 4A

OS LAB 03

BSSE SPRING 2026

LAB 03

1. `echo -e "Shah Faisal \n 22 \n Black "`



```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
Shah Faisal  
22  
Black  
lab@rawal-lab1-pc-01:~$
```

2. `#!/bin/bash`
`num=10`

```
if [ $num -ge 0 ]; then  
    echo "Positive"  
else  
    echo "Negative"  
fi
```

```
lab@rawal-lab1-pc-01:~$ ./script.sh
Shah Faisal
22
Black
lab@rawal-lab1-pc-01:~$ ./script.sh
./script.sh: line 2: [: -lt: unary operator expected
number is positive
lab@rawal-lab1-pc-01:~$ ./script.sh
Positive
lab@rawal-lab1-pc-01:~$
```

3.#!/bin/bash

num=15

if [\$num -lt 10]; then

 echo "Small"

elif [\$num -le 20]; then

 echo "Medium"

else

 echo "Large"

fi

```
lab@rawal-lab1-pc-01:~$ ./script.sh
Shah Faisal
22
Black
lab@rawal-lab1-pc-01:~$ ./script.sh
./script.sh: line 2: [: -lt: unary operator expected
number is positive
lab@rawal-lab1-pc-01:~$ ./script.sh
Positive
lab@rawal-lab1-pc-01:~$ ./script.sh
Medium
lab@rawal-lab1-pc-01:~$
```

```
4.for i in {1..5}
do
echo $i
done
```

```
lab@rawal-lab1-pc-01:~$ ./script.sh
Shah Faisal
22
Black
lab@rawal-lab1-pc-01:~$ ./script.sh
./script.sh: line 2: [: -lt: unary operator expected
number is positive
lab@rawal-lab1-pc-01:~$ ./script.sh
Positive
lab@rawal-lab1-pc-01:~$ ./script.sh
Medium
lab@rawal-lab1-pc-01:~$ ./script.sh
1
2
3
4
5
lab@rawal-lab1-pc-01:~$
```

```
5.count=1
while [ $count -le 5 ]
do
echo $count
count=$((count +1))
done
```

```
number is positive
lab@rawal-lab1-pc-01:~$ ./script.sh
Positive
lab@rawal-lab1-pc-01:~$ ./script.sh
Medium
lab@rawal-lab1-pc-01:~$ ./script.sh
1
2
3
4
5
lab@rawal-lab1-pc-01:~$ ./script.sh
./script.sh: line 1: i: command not found
lab@rawal-lab1-pc-01:~$ ./script.sh
./script.sh: line 2: [1: command not found
lab@rawal-lab1-pc-01:~$ ./script.sh
./script.sh: line 2: [1: command not found
lab@rawal-lab1-pc-01:~$ ./script.sh
1
2
3
4
5
lab@rawal-lab1-pc-01:~$
```

6.a=1

b=2

sum=0

sum=\$((a+b))

echo "\$sum"

```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
3  
lab@rawal-lab1-pc-01:~$
```

```
7.str=""  
if [ -n "$str" ]; then  
    echo "String is empty"  
else  
    echo "String is not empty"  
fi
```

```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
3  
lab@rawal-lab1-pc-01:~$ ./script.sh  
String is not empty  
lab@rawal-lab1-pc-01:~$ ./script.sh  
String is empty  
lab@rawal-lab1-pc-01:~$
```

```
8.#!/bin/bash
```

```
file="/home/lab/sohaib.txt"
```

```
if [ -f "$file" ] && [ -r "$file" ]; then
```

```
    echo "File exists and is readable"
```

```
else
```

```
    echo "File not readable"
```

```
fi
```

```
lab@rawal-lab1-pc-01:~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
=: cannot open `=' (No such file or directory)  
/home/lab/sohaib.txt: empty  
Condition not met  
lab@rawal-lab1-pc-01:~$ chmod 777 sohaib.txt  
lab@rawal-lab1-pc-01:~$ ./script.sh  
=: cannot open `=' (No such file or directory)  
/home/lab/sohaib.txt: empty  
Condition not met  
lab@rawal-lab1-pc-01:~$ ./script.sh  
=: cannot open `=' (No such file or directory)  
/home/lab/sohaib.txt: empty  
File not readable  
lab@rawal-lab1-pc-01:~$ ./script.sh  
=: cannot open `=' (No such file or directory)  
sohaib.txt: empty  
File not readable  
lab@rawal-lab1-pc-01:~$ ./script.sh  
=: cannot open `=' (No such file or directory)  
/home/lab/sohaib.txt: ASCII text  
File not readable  
lab@rawal-lab1-pc-01:~$ ./script.sh  
File exists and is readable  
lab@rawal-lab1-pc-01:~$
```

9.file="/home/lab/sohaib.txt"

```
if [ -f "$file" ] && [ -w "$file" ]; then  
    echo "File exists and is writable"  
else  
    echo "Condition not met"  
fi
```

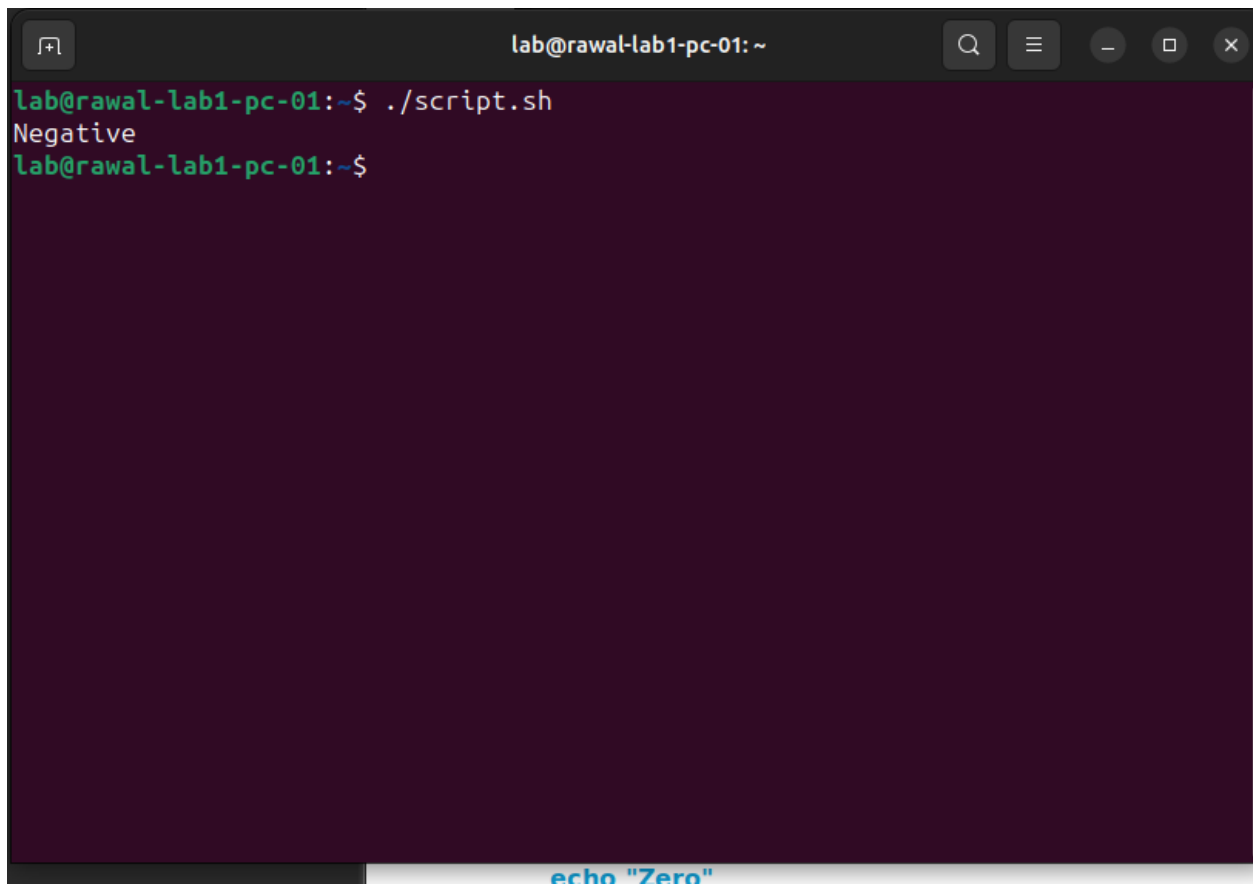
```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ chmod 777 sohaib.txt  
lab@rawal-lab1-pc-01:~$ ./script.sh  
=: cannot open `=' (No such file or directory)  
/home/lab/sohaib.txt: empty  
Condition not met  
lab@rawal-lab1-pc-01:~$ ./script.sh  
=: cannot open `=' (No such file or directory)  
/home/lab/sohaib.txt: empty  
File not readable  
lab@rawal-lab1-pc-01:~$ ./script.sh  
=: cannot open `=' (No such file or directory)  
sohaib.txt: empty  
File not readable  
lab@rawal-lab1-pc-01:~$ ./script.sh  
=: cannot open `=' (No such file or directory)  
/home/lab/sohaib.txt: ASCII text  
File not readable  
lab@rawal-lab1-pc-01:~$ ./script.sh  
File exists and is readable  
lab@rawal-lab1-pc-01:~$ ./script.sh  
Condition not met  
lab@rawal-lab1-pc-01:~$ ./script.sh  
File exists and is writable  
lab@rawal-lab1-pc-01:~$
```

10.dir="/home/lab/task2"

```
if [ ! -d "$dir" ]; then  
    echo "Directory does not exist"  
else  
    echo "Directory exist"  
fi
```

```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
Directory exist  
lab@rawal-lab1-pc-01:~$
```

```
11.#!/bin/bash  
num1=-15  
if [ $num1 -lt 0 ]; then  
    echo "Negative"  
elif [ $num1 -eq 0 ]; then  
    echo "Zero"  
elif [ $num1 -le 10 ]; then  
    echo "Small Positive"  
else  
    echo "Large Positive"  
fi
```



A terminal window titled 'lab@rawal-lab1-pc-01: ~' with a dark background. The prompt is 'lab@rawal-lab1-pc-01:~\$'. The user enters './script.sh'. The output is 'Negative'. The prompt returns to 'lab@rawal-lab1-pc-01:~\$'. At the bottom of the terminal, the text 'echo "Zero"' is visible in a light blue font.

```
lab@rawal-lab1-pc-01:~$ ./script.sh
Negative
lab@rawal-lab1-pc-01:~$
echo "Zero"
```

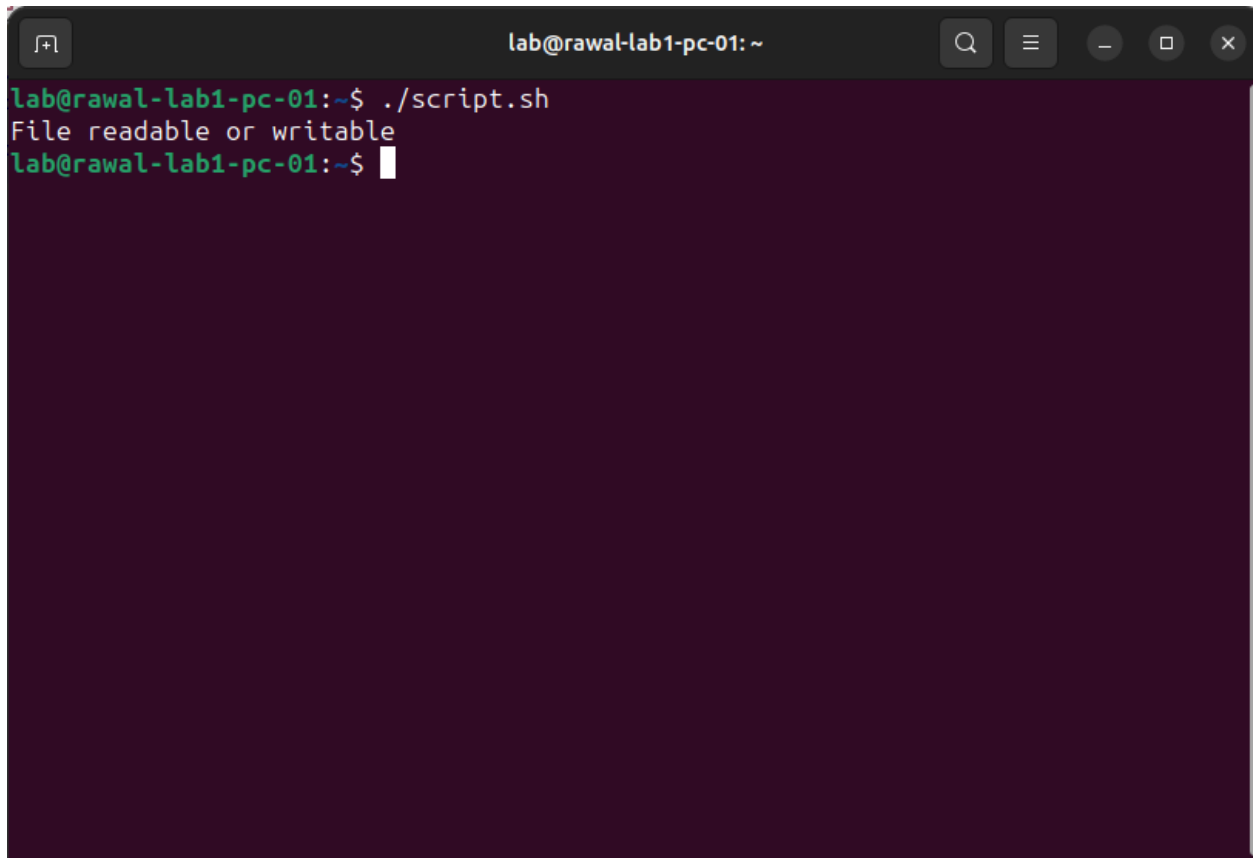
```
12.file="/home/lab/sohaib.txt"
```

```
if [ -r "$file" ] && [ -x "$file" ]; then
    echo "File readable and is exectuable"
else
    echo "Condition not met"
fi
```

```
e" 1 2 2 [ x "$file" ] ; then
f
lab@rawal-lab1-pc-01: ~
lab@rawal-lab1-pc-01:~$ ./script.sh
Directory exist
lab@rawal-lab1-pc-01:~$ ./script.sh
File readable and is exectuable
lab@rawal-lab1-pc-01:~$
```

13.file="/home/lab/sohaib.txt"

```
if [ -r "$file" ] || [ -w "$file" ]; then
    echo "File readable or writable"
else
    echo "Condition not met"
fi
```



```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
File readable or writable  
lab@rawal-lab1-pc-01:~$
```

```
14.for i in {1..10}  
do  
    if [  $((i\%2))$  -eq 0 ]; then  
        echo "$i is Even"  
    else  
        echo "$i is Odd"  
    fi  
done
```

```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
File readable or writable  
lab@rawal-lab1-pc-01:~$ ./script.sh  
1 is Odd  
2 is Even  
3 is Odd  
4 is Even  
5 is Odd  
6 is Even  
7 is Odd  
8 is Even  
9 is Odd  
10 is Even  
lab@rawal-lab1-pc-01:~$
```

```
15.i=1  
while [ $i -le 10 ]  
do  
    if [ $i -eq 5 ]; then  
        break  
    fi  
    echo $i  
    i=$((i+1))  
done
```

```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
1  
2  
3  
4  
lab@rawal-lab1-pc-01:~$
```

```
16.dir="/home/lab/task2"  
file="/home/lab/sohaib.txt"
```

```
if [ -d "$dir" -a -w "$dir" ] -o [ -f "$file" -a -r "$file" ]; then  
    echo "Condition satisfied"
```

```
else  
    echo "Condition not satisfied"
```

```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
./script.sh: line 4: [: too many arguments  
Condition not satisfied  
lab@rawal-lab1-pc-01:~$
```

```
lab@rawal-lab1-pc-01:~$ ./script.sh  
./script.sh: line 4: [: too many arguments  
Condition not satisfied  
lab@rawal-lab1-pc-01:~$
```

```
fi
```

```
lab@rawal-lab1-pc-01:~$ ./script.sh  
./script.sh: line 4: [: too many arguments  
Condition not satisfied  
lab@rawal-lab1-pc-01:~$
```

```
17.for i in  
   {1..10}  
do  
    if [ $i -eq 7  
]; then  
        continue  
    fi  
    echo $i  
done
```

```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
1  
2  
3  
4  
5  
6  
8  
9  
10  
lab@rawal-lab1-pc-01:~$
```

18.while true

do

 echo "Running..."

 read -p "Type stop to exit: " input

 if ["\$input" = "stop"]; then

 break

 fi

done



```
lab@rawal-lab1-pc-01: ~  
lab@rawal-lab1-pc-01:~$ ./script.sh  
Running...  
Type 5 to exit:  
Running...  
Type 5 to exit:  
Running...  
Type 5 to exit: da  
Running...  
Type 5 to exit: 1  
Running...  
Type 5 to exit: 5  
lab@rawal-lab1-pc-01:~$
```